

Gene Expression Profiles at Early vs Late Stages After Cervical Artery Dissection

Robert B. Ferguson,^{1,*} Ilana E. Green,^{2,*} Timothy L. Mcmurry,³ Keith L. Keene,⁴ Stephen R. Williams,³ Fang-Chi Hsu,⁵ Stefan Bekiranov,⁶ Hyacinth I. Hyacinth,⁷ Glen C. Jickling,⁸ Michèle M. Sale,³ Bradford B. Worrall,^{1,3} and Andrew M. Southerland^{1,3}

Correspondence
Dr. Southerland
as5ef@virginia.edu

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Abstract

Background and Objectives

Cervical artery dissection (CeAD) is a common cause of stroke in young adults. While the pathogenesis of CeAD remains poorly understood, clinical and genetic associations suggest a systemic arteriopathy at the early phase of CeAD. We hypothesize that individuals experiencing CeAD express a distinct gene expression profile at the early phase compared with (1) age-matched and sex-matched controls and (2) late-phase profiles within the same individual at a later time point.

Methods

We prospectively enrolled patients (n = 37) with CeAD with and without ischemic stroke (IS) from 2014 to 2016, excluding cases of significant trauma or known monogenic connective tissue disease. We compared CeAD cases with non-CeAD IS cases (n = 16) and healthy controls (n = 11). Study personnel collected blood samples within 4 weeks of the event and again at a 3–6-month follow-up. We assessed gene expression using microarray analysis at the time of CeAD, relative to controls and >3-month follow-up samples within CeAD cases. We applied a mixed-effects regression model that included covariates of age, sex, race/ethnicity, time of enrollment, and stroke occurrence. We used a false discovery rate (FDR) cutoff of 5% and a 1.5-fold change cutoff to identify robust differential expression.

Results

Within early-phase CeAD cases, 1,238 genes showed differential expression at a 5% FDR compared with late-phase profiles, with 31 genes exhibiting at least a 1.5-fold change. We identified 538 differentially expressed genes between patients with CeAD and controls, of which 30 reached the 1.5-fold threshold. Comparison of late-phase profiles with controls revealed 11 differentially expressed genes. Gene ontology highlighted that 7 of the 11 genes were associated with hemoglobin metabolism.

Discussion

In this preliminary gene expression analysis of patients with CeAD, we identified a distinct gene expression profile associated with early-phase dissection. The predominance of differential gene expression related to hemoglobin metabolism warrants further study and replication in independent cohorts of patients with arterial dissection.

Introduction

Stroke is a leading cause of death and disability worldwide.¹ Stroke in young adults is particularly devastating, given both the individual and societal impact from the greater number of quality-adjusted life years lost to disability. Cervical artery dissection (CeAD)—dissection of

*These authors contributed equally to this work as co-first authors.

¹Department of Neurology, University of Virginia, Charlottesville; ²Department of Neurology, Albert Einstein College of Medicine, New York, NY; ³Department of Public Health Sciences, University of Virginia, Charlottesville; ⁴Department of Biology, East Carolina University, Greenville, NC; ⁵Department of Biostatistics and Data Science, Wake Forest University School of Medicine, Winston-Salem, NC; ⁶Department of Biochemistry and Molecular Genetics, University of Virginia, Charlottesville; ⁷Department of Neurology, University of Cincinnati, OH; and ⁸Department of Medicine, University of Alberta, Edmonton, Canada.

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Glossary

CeAD = cervical artery dissection; **cirRNA** = circular RNA; **FDR** = false discovery rate; **FMD** = fibromuscular dysplasia; **IS** = ischemic stroke; **UVA** = University of Virginia.

the carotid or vertebral arteries—is a significant cause of ischemic stroke (IS) in young adults, accounting for nearly 1 in every 5 cases of stroke in adults younger than 55 years.²⁻⁴

Most cases are seemingly spontaneous, although minor neck trauma such as whiplash or chiropractic manipulation is a frequent association. Another subset of CeAD is associated with major neck trauma, such as motor vehicle accidents, and is likely distinct from nontraumatic cases.⁵⁻⁷ A minute proportion of nontraumatic CeAD (<2%) can be attributed to known monogenic connective tissue disorders, such as vascular Ehlers-Danlos or Marfan syndrome.⁸⁻¹⁰ Another 10%–20% of spontaneous cases occur in patients with fibromuscular dysplasia (FMD), a nonatherosclerotic arteriopathy of unclear etiology resulting in weakness of the arterial media. FMD shares a genetic association with CeAD through the recently identified PHACTR1 risk allele rs9349379.^{11,12}

Similar to other cerebrovascular events such as intracranial aneurysms,¹³ arteriovenous malformations,¹⁴ and IS,¹⁵ we hypothesize that CeAD may have a distinct gene expression signature within the early phase of the event, consistent with a systemic and potentially inflammatory transient arterial vulnerability. Microarray-based gene expression analysis is a method for investigating which genes are expressed in specific cells or tissues at a given time, providing a bridge to understanding the relationship between genetic risk and environmental factors.

In this preliminary analysis, we tested the hypotheses that patients experiencing CeAD (1) express a distinct gene expression signature at the early phase compared with those with ischemic strokes, not due to dissection and with healthy controls and (2) express a distinct gene signature at the early phase of the CeAD event compared with their late-phase profiles at 3–6 months after the event.

Methods

Study Population

The following criteria determined enrollment eligibility: CeAD—carotid or vertebral artery dissection occurring within 2 weeks of clinical diagnosis determined by a board-certified vascular neurologist and confirmatory cervical vascular imaging including any of the following modalities: CT angiography of the neck, MRI, and angiography of the neck (T1 fat-saturated sequences), or catheter-based cerebrocervical angiography; IS (CeAD cases and controls including non-CeAD stroke cases and healthy controls)—a focal neurologic deficit localizing to a distinct cerebrovascular territory,

confirmed by MRI of the brain with diffusion-weighted imaging or head CT with well-defined cerebral infarction.

Exclusion criteria were as follows: (1) CeAD cases associated with significant trauma—defined as trauma requiring hospital admission and/or surgical or medical intervention independent of dissection; (2) iatrogenic CeAD; (3) pure intracranial artery dissection; (4) CeAD originating from aortic dissection; (5) known monogenic connective tissue disorder; (6) evidence of cervical or cephalic vasculopathy not related to dissection (other than FMD); (7) known pregnancy; (8) patients unable to complete anticipated 3–6 months of follow-up; (9) patients unable to provide informed consent and without a legally authorized representative; and (10) age <18. For healthy controls, we excluded anyone with a history of stroke, CeAD, or active inflammatory disease.

Data Collection

We prospectively enrolled participants from consecutive admissions to the inpatient Vascular Neurology service and outpatient clinics at the University of Virginia (UVA) Medical Center from 2014 to 2016. Ischemic stroke controls (no CeAD) were also recruited prospectively from the inpatient Vascular Neurology service, the outpatient Stroke Clinic, and the Neurology consultation service. Healthy controls were recruited from (1) healthy spouses and unrelated friends of cases and stroke controls in the inpatient setting, (2) healthy spouses and friends of stroke patients in the outpatient clinics, and (3) healthy volunteers solicited through local media.¹⁶

All participants underwent whole-blood venous sampling at the time of enrollment, within 4 weeks of the diagnosis of CeAD, and again at 3–6 months for CeAD and non-CeAD stroke cases. Samples were collected from each consenting participant into RNA PAXgene tubes (Qiagen Inc.) and stored at –80°C. Laboratory personnel at the UVA Center for Public Health Genomics then processed the samples in a core laboratory. RNA was isolated from blood leukocytes, followed by reverse transcription, amplification (Ambion RNA amplification kit), sample labeling, and microarray analysis.

All clinical data were collected and managed using REDCap, an electronic data capture system.¹⁷ For internal quality control, medical records were reviewed for all CeAD and non-CeAD participants to validate and supplement patient-reported data.

Microarray Analysis

Samples were analyzed using Illumina HumanHT-12 V4.0 beadchips,¹⁸ each of which contains wells for 12 samples.^{15,19,20} The research team randomly assigned samples to

beadchips to ensure the representation of each sample class on every chip and to avoid placing multiple samples from the same individual on a single beadchip. We used the remaining wells for randomly chosen technical replicates and assigned samples to wells within each beadchip.

Statistical Analysis

Data were preprocessed (i.e., background-corrected, variance-stabilized, and quantile-normalized) and assessed for quality control using the standard R lumi Bioconductor package workflow.²¹ The analysis retained all probes matching gene symbols in the Bioconductor illuminaHumanv4.db database. We evaluated expression-level differences using mixed-effects models for each retained probe.²² To account for the differential occurrence of stroke within patients with CeAD, probe-level expression was modeled depending on the time of CeAD as well as the time of stroke. The time of CeAD was estimated based on patients' initial reports of symptoms, including neck pain, headache, or ptosis associated with Horner syndrome. The models further controlled for age, sex, and race/ethnicity and used random effects to account for repeated measures on patients and sample clustering within beadchips.

Differences in gene expression at the time of CeAD, controlling for stroke or nonstroke, were assessed using the coefficient for the baseline CeAD indicator. We assessed longitudinal changes in gene expression by comparing baseline expression with the average gene expression at the 3–6-month sampling point. Selected genes were considered significant at a 5% false discovery rate (FDR) and with at least a 1.5-fold change in expression.

We present patient characteristics as means and standard deviations for continuous variables, medians for continuous variables, and counts and percentages for categorical variables. Table 1 summarizes baseline cohort characteristics and is presented for descriptive purposes only, and no formal statistical comparisons were performed across groups for these variables.

Clustering Analysis

We selected genes with an FDR < 0.05 and the highest average expression differences between cases and controls for hierarchical clustering. Heat maps were generated based on z-score-normalized gene expression across samples (i.e., row normalized), and technical replicates were averaged prior to normalization. Genes were hierarchically clustered according to their expression levels using "R's hclust" (hierarchical clustering) function with a Euclidean distance metric and complete linkage.

Standard Protocol Approvals, Registrations, and Patient Consents

All study procedures followed a protocol that was reviewed and approved by the UVA Institutional Review Board. All study investigators completed the Collaborative Institutional Training Initiative certification, as required by ethical standards for human subject research. Written informed consent

was obtained from all participants or their legally authorized representatives prior to enrollment.

Data Availability

The data that support the findings of this study are not publicly available because of ethical and privacy considerations but may be made available from the corresponding author on reasonable request.

Results

Patient Demographics

Table 1 presents patient demographics. The median time from the early phase of CeAD to first blood draw was 6.5 days. There was no difference in age, sex, race/ethnicity, or presence of hypertension, hyperlipidemia, or migraine between the CeAD and non-CeAD IS or healthy control groups. Patients in the non-CeAD IS group were more likely to have diabetes and to smoke than patients in the CeAD group ($p < 0.01$).

Gene Expression Analysis

In the control comparison, we identified 538 differentially expressed genes between patients with CeAD and a combined group of healthy controls and non-CeAD IS participants, with 30 transcripts exhibiting at least 1.5-fold change (Figure 1A). In the late-phase comparison, 1,238 genes were differentially expressed between early-phase CeAD and late-phase profiles within participants at 3–6 months, with 31 of these transcripts reaching the 1.5-fold change difference (Figure 1B). Of the gene transcripts that showed a 1.5-fold change between the control and late-phase comparisons, 11 genes overlapped in analyses of early-phase vs late-phase and early-phase vs the control groups. Log-transformed expression and directionality of these 11 genes are illustrated in Figure 1. Of the 11 genes that were most differentially expressed in the early phase of CeAD, 7 were involved in hemoglobin metabolism, with alpha hemoglobin stabilizing protein (AHSP) showing the most differential expression (Table 2).

Heat maps illustrate a pattern of hierarchical clustering and differential gene expression between early-phase CeAD (with or without stroke), healthy controls (Figure 2), non-CeAD IS controls (Figure 3), and late-phase profiles (Figure 4).

Discussion

This gene expression study and early-phase analysis in CeAD revealed a unique profile of 11 genes that are differentially expressed in patients with CeAD at the early phase compared with the late-phase samples and non-CeAD controls. These findings support the hypothesis that there may be a unique and transient signature of gene expression that occurs around the time of CeAD.

There are likely ephemeral environmental and physiologic factors associated with incidence of CeAD. Observational

Table 1 Patient Characteristics

	CeAD (n = 36)*	Non-CeAD stroke (n = 16)	Healthy** (n = 11)
Mean age $\pm$ SD (y)	41 $\pm$ 12	44 $\pm$ 9	39 $\pm$ 13
Sex (% female)	44	56	55
Race (%)***	80 White 14 Black 6 Asian	69 White 31 Black	100 White
Hypertension (%)	28	75	14
Hyperlipidemia (%)	25	69	0
Diabetes (%)	8	25	0
Migraine (%)	28	44	27
Smoking (%)	11	56	14

Abbreviation: CeAD = cervical artery dissection.

Table 1 is presented for descriptive purposes only, and no formal statistical testing was performed for comparisons among groups: *69% of CeAD cases presenting with stroke; **missing data were not imputed; ***race self-identified.

studies reveal that CeAD peaks in occurrence at a mean age range of 42–44 years, with 75% of cases occurring from ages 30 to 50.⁸ After an early-phase CeAD event, the risk of recurrent dissection is highest in the first few weeks ($\sim$ 25%), while the long-term risk of recurrence ($>$ 3 months) is low ($\sim$ 1%–2% per year).²³ In addition, CeAD is likely a consequence of a generalized arteriopathy, given that 20% of cases involve multivessel dissection²⁴ and histologic data demonstrate abnormal ultrastructure in clinically unaffected arteries.^{25,26}

In addition, transient inflammatory factors may be associated with CeAD. Several retrospective studies show the association of CeAD with infection, particularly upper respiratory infections, although cervical exertion from violent coughing or retching may be a confounding factor.^{27,28} Elevated inflammatory markers, including c-reactive protein and absolute leukocyte count, are also observed in patients with CeAD. Notably, a retrospective analysis reported higher levels of these inflammatory markers in spontaneous CeAD cases compared with traumatic cases.^{29,30} Imaging studies of CeAD using high-resolution MRI and/or

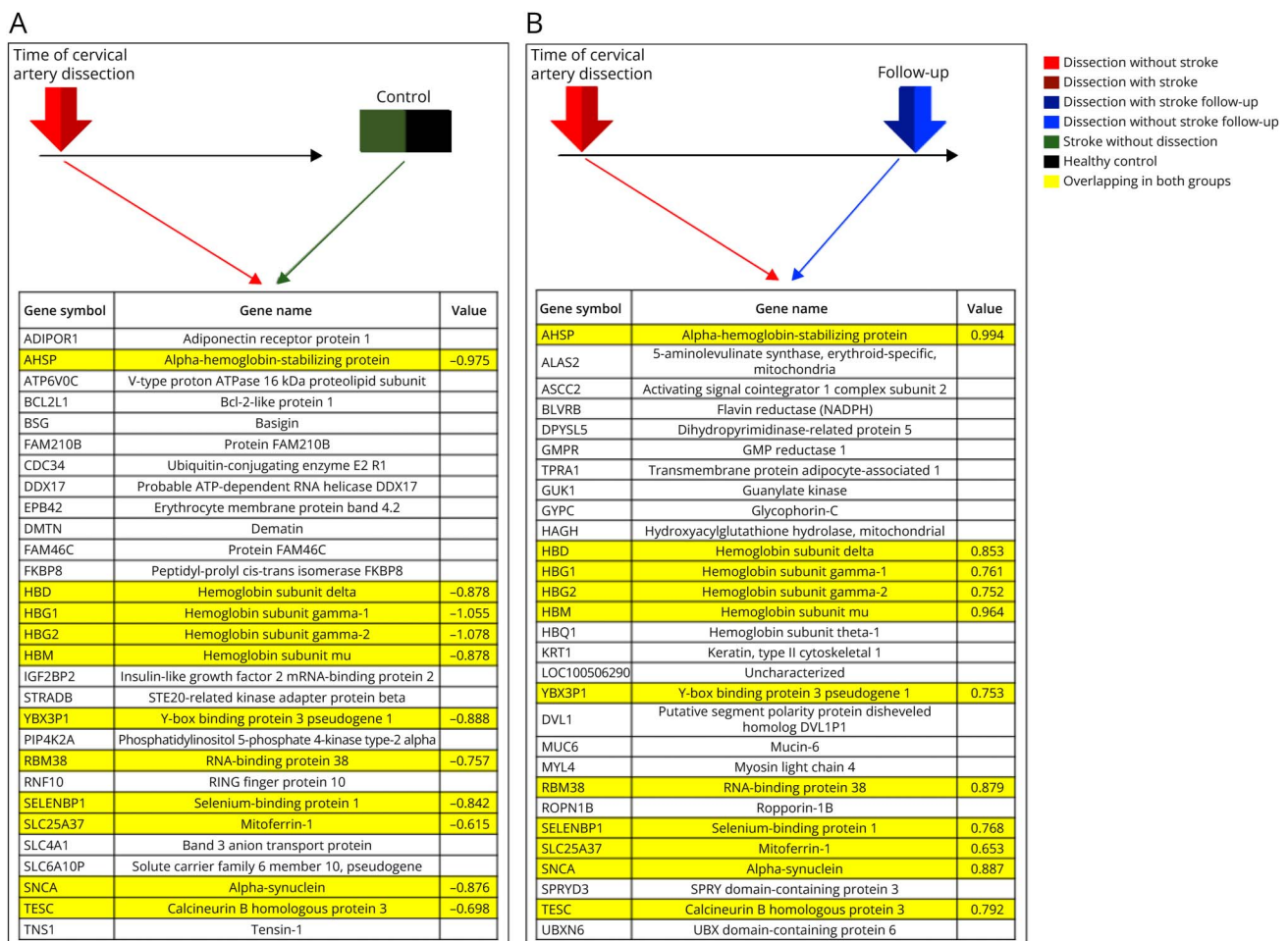
Figure 1 Log2 Fold Change of Overlapping Genes Showing Directionality for (A) Early-Phase Cervical Artery Dissection (CeAD) vs Non-CeAD Stroke and Healthy Controls and (B) Early-Phase vs Late-Phase CeAD Profiles at 3–6 months

Table 2 Differentially Expressed Genes in CeAD Between Early-Phase and Late-Phase Profiles and Between Early-Phase CeAD and Stroke Patients/Healthy Controls, Related to Hemoglobin Metabolism

Gene	Relation to hemoglobin metabolism	Log ₂ fold changes (A)	Log ₂ fold changes (B)
HBG1	Produce gamma subunits of fetal Hgb, <1% Hgb in adults; higher in promotor variant, delta-beta thalassemia, and sickle cell anemia	−1.055	0.761
HBG2		−1.078	0.752
HBD	Produce delta subunits of Hgb A2; <2% Hgb in adults; variants cause delta-thalassemia	−0.878	0.853
AHSP	Alpha hemoglobin stabilizing protein; chaperones alpha globin for transfer to beta globin to make tetrameric Hgb (hemoglobin A); plays a role in thalassemias and HGH disease	−0.975	0.994
SLC25A37	Also known as mitoferrin-1, a mitochondrial iron importer essential to heme biosynthesis and erythroid development	−0.615	0.653
RBM38	Expressed in spleen and bone marrow, critical to erythroid differentiation	−0.757	0.879
SNCA	Expression regulated by iron metabolism	−0.876	0.887

Abbreviations: AHSP = alpha hemoglobin stabilizing protein; HBD = hemoglobin subunit delta; HBG = hemoglobin subunit gamma; RBM = RNA-binding motif; SNCA = synuclein alpha; SLC = solute carrier family (Mitoferrin-1). Log₂ fold change (A) represents differential expression in participants with cervical artery dissection (CeAD), with and without ischemic stroke, compared with non-CeAD stroke controls and healthy controls. Log₂ fold change (B) represents within-participant longitudinal comparison of early-phase vs late-phase gene expression profiles in participants with CeAD (with and without stroke) at the 3–6-mo follow-up sampling.^{43,44}

positron emission tomography-computed tomography demonstrate arterial wall changes in the cervical arteries purportedly corresponding to diffuse vessel wall inflammation, even in clinically unaffected arteries, with subsequent normalization of these vessel wall changes at 6 months after dissection.^{31,32} It is important to note that, however, histopathology from case reports and case series of CeAD fails to demonstrate an inflammatory infiltrate in the affected arteries.^{25,26,33}

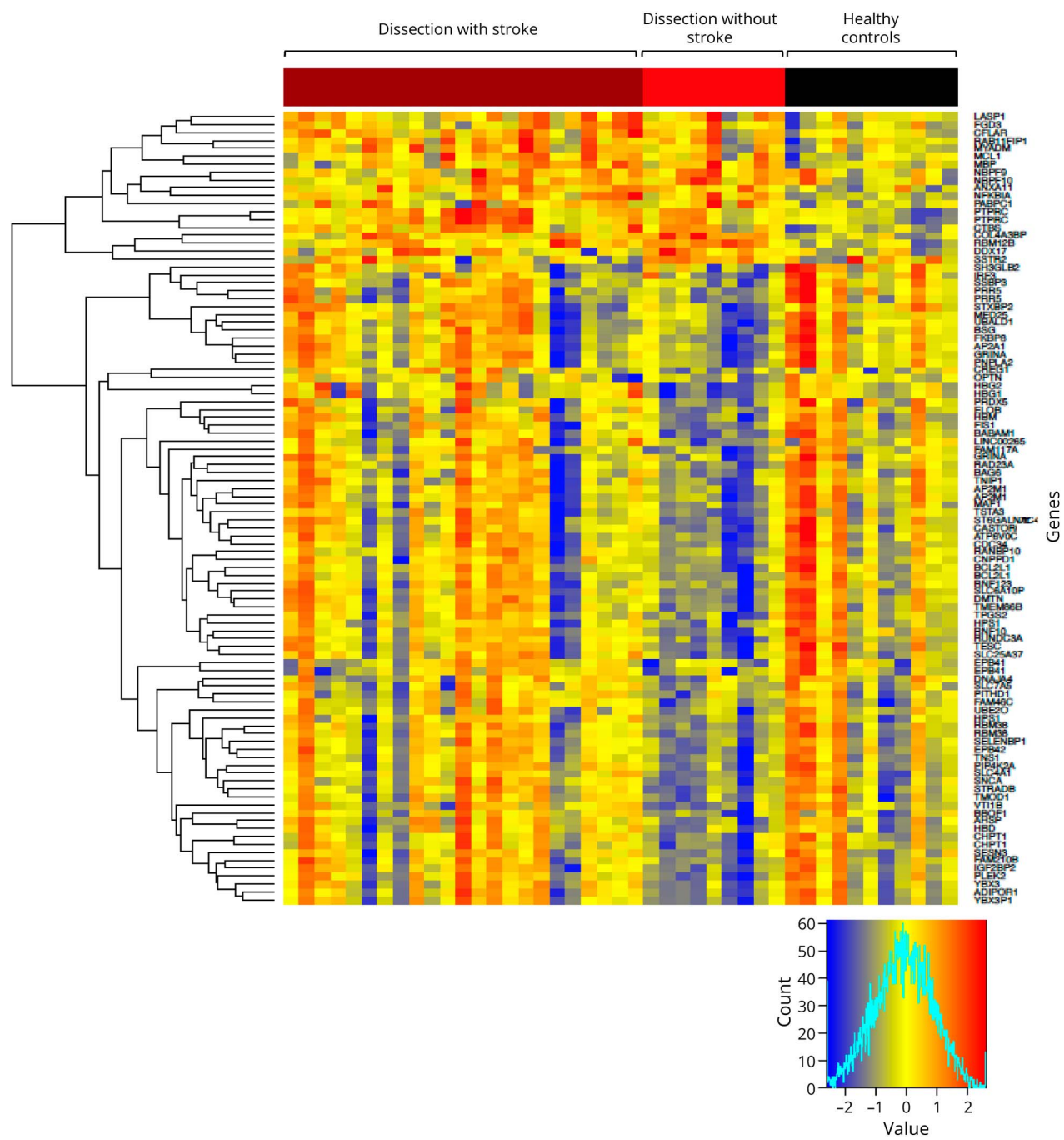
Previous and ongoing genetic research in CeAD has focused on candidate gene analyses and genome-wide association studies.^{10,24,34–36} Although that research is important for discovering the underlying genetic determinants of structural arterial weakening in the causal pathway for CeAD, it cannot account for the temporality of risk and age-related vulnerability in the pathogenesis. A prior study in patients with CeAD examined peripheral blood RNA sequencing, identifying gene expression changes associated with mitochondrial dysfunction and vascular remodeling. The study had its limitations due to the vast enrollment window from the time of the event, 12 months, and the lack of longitudinal assessment of its study population.³⁷ Another study explored the expression of circular RNA (circRNA) in patients with CeAD and identified differentially expressed circRNAs involved in transcriptional and post-transcriptional regulation.³⁸ Relatedly, a genome-wide association meta-analysis of patients with spontaneous coronary artery dissection revealed genetic variants associated with arterial integrity and tissue-mediated coagulation, highlighting shared pathophysiologic pathways with CeAD.³⁹ Collectively, these observational studies suggest the potential of genomic approaches to better understand the complex pathogenesis of CeAD.

In our study, 7 of the 11 genes that were most differentially expressed in the early phase of CeAD compared with controls

and late-phase samples are involved in hemoglobin metabolism, with AHSP demonstrating the greatest differential expression (Table 2). This finding generates the hypothesis that vulnerability to CeAD or occurrence of dissection in other arteries (e.g., coronary and aorta) could relate to oxygen transport. Impaired oxygen delivery to the vasa vasorum supplying the arterial media may cause capillary defects and trigger neoangiogenesis in the vessel wall, which can predispose patients to spontaneous intramural hemorrhage—a finding described in temporal artery biopsies of individuals with CeAD.^{28,29} Alternatively, arterial dissection could relate to hemoglobin function, independent of oxygen-carrying capacity. For instance, hemoglobin alpha is expressed in endothelial cells and modulates nitric oxide signaling with smooth muscle cells, affecting blood flow, arterial tone, and blood pressure, which is associated with CeAD.^{40–42} Hemoglobin gene expression may also play a role in systemic inflammatory response. Both hemoglobin alpha and beta are expressed in mucosal surfaces, exhibiting antimicrobial and antioxidative activities. Leukocytes also express hemoglobin proteins despite having no role in oxygen transport. Alternatively, given the inability to obtain whole blood samples prior to dissection, these observed gene expression differences could be a physiologic response to the vessel wall injury rather than a causal factor.

While these data generate additional hypotheses, the small sample size, single-center analysis, variability in time to blood draw, and lack of diversity in healthy controls constrain the generalizability of the conclusion. Given the small sample size, we were unable to account for potential confounders, such as comorbidities, stroke severity, and infarct size in the stroke group, as well as characteristics of the dissection itself, including location and degree of stenosis or occlusion. Moreover, factors such as medication usage and detailed stratification of stroke subtype were not accounted for and

Figure 2 Heat Map of Differentially Expressed Genes (FDR <0.05) Comparing Early-Phase CeAD (With and Without Stroke) and Healthy Controls

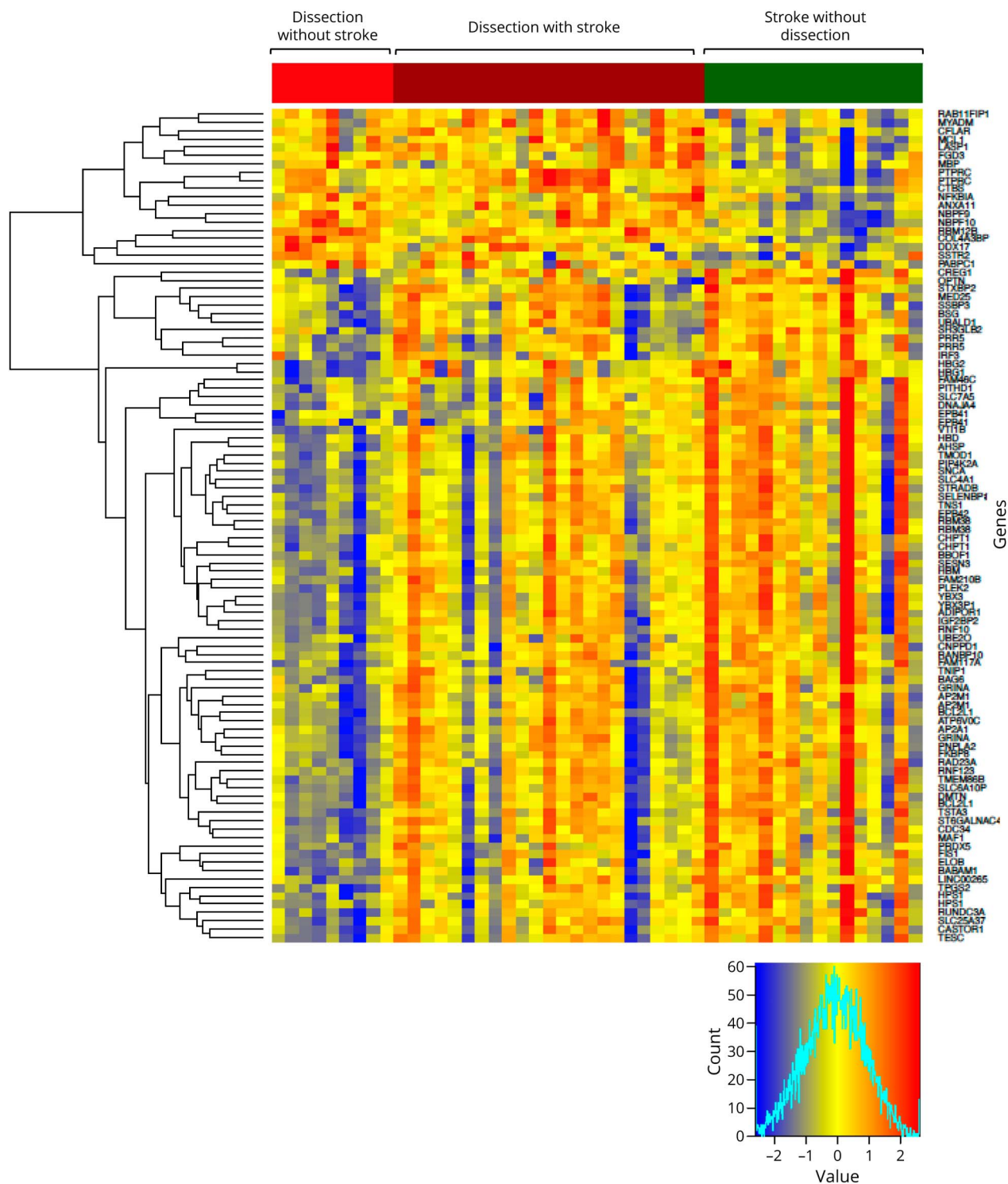


may have potentially influenced the observed gene expression findings. Samples from participants' spouses may also reflect environmental associations unaccounted for in the analysis. In addition, the variable interval between CeAD diagnosis and blood sample collection may have influenced the observed gene expression profiles and should be considered both when interpreting these data and in the design for future study. Convenience or consecutive sampling may have introduced selection bias by excluding more severe cases and contributing to early mortality. Future studies could include sampling from

vessel wall connective tissue, such as temporal artery or skin biopsy, in addition to whole-blood sampling.

In this study of RNA expression in patients with CeAD, we identified a distinct gene expression profile at the early phase of CeAD compared with the late-phase CeAD samples and non-CeAD controls. Further research is needed to investigate the potential role of hemoglobin metabolism in the pathophysiology of CeAD. Although these findings are hypothesis-generating, future analyses are warranted to explore gene

Figure 3 Heat Map of Differentially Expressed Genes (FDR <0.05) Comparing Early-Phase CeAD and Non-CeAD Ischemic Stroke Controls



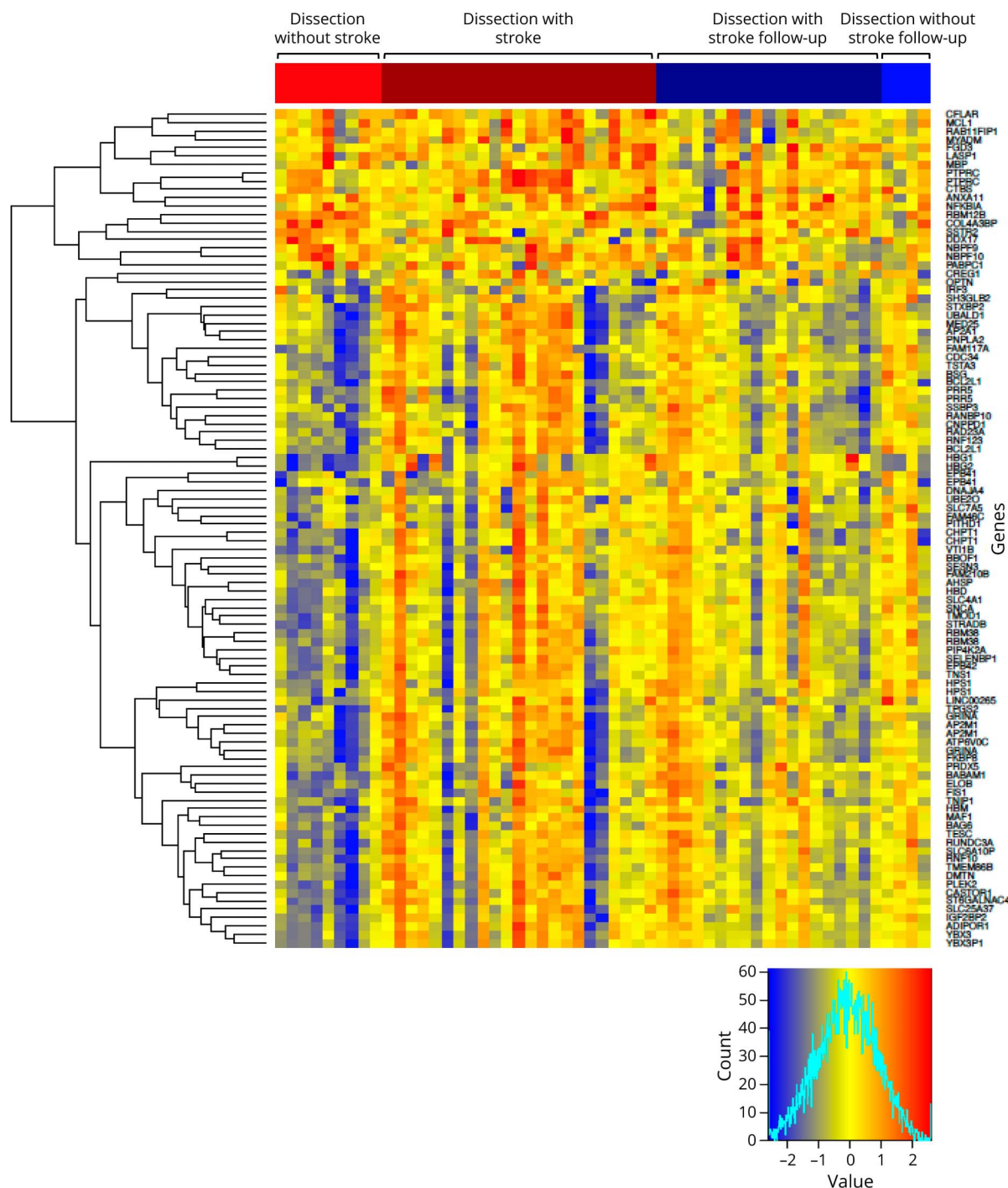
expression signatures of CeAD in larger, more diverse, and independent cohorts.

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Figure 4 Heat Map of Differentially Expressed Genes (FDR <0.05) Comparing Early-Phase and Late-Phase CeAD Profiles at 3–6 months



Author Contributions

R.B. Ferguson: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. I.E. Green: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. T.L. Mcmurry: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data.

K.L. Keene: drafting/revision of the manuscript for content, including medical writing for content. S.R. Williams: drafting/revision of the manuscript for content, including medical writing for content. F-C. Hsu: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. S. Bekiranov: drafting/revision of the manuscript for content, including medical writing for content; major role in the

acquisition of data; analysis or interpretation of data. H.I. Hyacinth: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. G.C. Jickling: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M.M. Sale: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. B.B. Worrall: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A.M. Southerland: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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Disclosure

A. Southerland reports research support from the American Heart Association/American Stroke Association; serves on editorial boards for Stroke, Journal of Stroke and Cerebrovascular Disease, and Frontiers in Stroke; and performs legal expert consultation related to CeAD. B. Worrall reports research support from the NIH and AHA/ASA and serves as Associate Editor for Editorial Education for Neurology. All other authors report no relevant disclosures. Go to Neurology.org/NG for full disclosures.

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